

Review

62%

8

Out of 13 points

1:29

Avg Time Per Question

Your Answers:

Vignette

The chief investment officer asks you to analyze one of the firm's portfolios to identify influential outliers that might be skewing regression results of its return drivers. For each observation, you calculate leverage and the studentized residual. There are 96 observations and two independent variables ($k = 2$), and the critical t -statistic is 2.63 at a 1% significance level. Partial results of your calculations are shown in Exhibit 1.

Exhibit 1

Regression Data for Detecting Influential Observations

	h_{ii}	Studentized Residual
Observation 1	0.043	2.784
Observation 2	0.022	-0.103
Observation 3	0.036	-0.731
Observation 4	0.059	-0.122
Observation 5	0.011	-0.660
Observation 6	0.101	-2.906
Observation 45	0.042	2.117

	h_{ij}	Studentized Residual
Observation 46	0.013	0.172
Observation 47	0.015	-0.672
Observation 48	0.012	-0.734
Observation 49	0.064	0.475
Observation 50	0.141	-2.788
Observation 51	0.011	1.679
Observation 52	0.023	-1.218
Observation 91	0.035	-1.260
Observation 92	0.025	3.001
Observation 93	0.017	1.483
Observation 94	0.097	-0.172
Observation 95	0.017	0.046
Observation 96	0.011	1.819

Finally, you are tasked with investigating whether there is any monthly seasonality in the excess portfolio returns. You construct a regression model using dummy variables for the months; your regression statistics and ANOVA results are shown in Exhibit 2.

Exhibit 2

Analysis of Monthly Seasonality of Excess Portfolio Returns

Regression Statistics

Multiple R	0.321
R -Squared	0.103
Adjusted R -Squared	-0.014

Regression Statistics

Standard Error	8.100
Observations	96.000

ANOVA

	df	SS	MS	F	Signif. F
Regression	11	634.679	57.698	0.879	0.563
Residual	84	5511.369	65.612		
Total	95	6146.048			

	Coeff.	Std. Error	t-Stat.	P-Value
Intercept	1.263	2.864	0.441	0.660
Jan	1.311	4.050	0.324	0.747
Feb	-3.756	4.050	-0.927	0.356
Mar	3.495	4.050	0.863	0.391
Apr	0.174	4.050	0.043	0.966
May	0.714	4.050	0.176	0.861
Jun	0.944	4.050	0.233	0.816
Jul	-0.571	4.050	-0.141	0.888
Aug	-0.445	4.050	-0.110	0.913
Sep	-1.744	4.050	-0.431	0.668
Oct	4.261	4.050	1.052	0.296
Nov	-5.311	4.050	-1.311	0.193

1 Multiple Choice 0 / 1 point

Determine and justify the potentially influential observation(s) in Exhibit 1 using the leverage measure.

☐ A. 50, because it has the highest leverage, 0.141



☒ B. Observations 6 and 50, because their leverage exceeds 0.100

Correct Answer:

C. Observations 6, 50, and 94, because their leverage exceeds _____



C. Observations 6, 50, and 94, because their leverage exceeds _____

Feedback

C is correct. The rule of thumb for the leverage measure is that if it exceeds

_____, where k is the number of independent variables, then it is a

potentially influential observation. Since $n = 96$ and $k = 2$, then

_____. Three observations exceed this value: 6, 50, and 94.

So, they are potentially influential observations.



2 Multiple Choice 1 / 1 point

Determine and justify the potentially influential observations in Exhibit 1 using the studentized residuals measure.



A. Observations 1 and 92, because the values of their studentized residuals exceed 2.63



☒ B. Observations 1, 6, 50, and 92, because the absolute values of their studentized residuals exceed 2.63



C. All the observations shown except Observation 95, because the absolute value of its studentized residual is less than 0.094

Feedback

B is correct. For the studentized residuals measure, the critical t -value is 2.63. So, any observation with a studentized residual whose absolute value exceeds 2.63 is a potentially influential observation. The studentized residuals for Observations 1, 6, 50, and 92 have absolute values exceeding 2.63; therefore, they are potentially influential observations.



3 Multiple Choice 1 / 1 point

Identify both the base month and the coefficient that represents its returns in Exhibit 2.



☒ A. December is the base month, and the intercept coefficient represents its returns.

- ☐ B. November is the base month, and the intercept coefficient represents its returns.
- ☐ C. December is the base month, and the average of the coefficients for the other 11 months represents its returns.

Feedback

A is correct. December is the base month, and the intercept coefficient represents its returns. We use 11 dummy variables to represent the returns for each month from January through November. December results are measured when each of these dummy variables equals zero, leaving the intercept coefficient to represent December returns.

4 Multiple Choice 0 / 1 point

Determine using Exhibit 2 which one of the following statements is *most likely* to be correct. Monthly seasonality in the firm's portfolio is:



☒ A. highly likely.

Correct Answer: **B. highly unlikely.**

- ☐ B. highly unlikely.
- ☐ C. not able to be determined from the given data.

Feedback

B is correct. Monthly seasonality in the firm's portfolio is highly unlikely. The variance explained by the model (R -squared) is only 10.3%, and after adjusting for the number of independent variables (adjusted R -squared), it becomes negative. Also, the insignificant F -statistic indicates a 56.3% chance that all variable coefficients are zero. Finally, t -statistics and associated p -values indicate that all the variable coefficients are insignificant (i.e., not significantly different from zero). Consequently, monthly seasonality is highly unlikely to exist in this portfolio.

Vignette

Your second-round interview for the Junior Quantitative Analyst position went well, and the next day, you receive an email from the investment firm congratulating you for making it

this far. You are one of four remaining candidates from more than 100 who applied for the position.

Because the position involves quantitative analysis, you are given an assignment to complete within 72 hours. You are provided a dataset and tasked with creating two logistic regression models to predict whether an exchange-traded fund (ETF) will be a “winning” fund—that is, whether the ETF’s monthly return will be one standard deviation or more above the mean monthly return across all ETFs in the dataset or whether the ETF will be an “average” fund.

The variables in the dataset are as follows:

	Variable	Description
1	net_assets	Net assets of ETF in US dollars
2	small_fund	Dummy variable = 1 if a small-size fund and 0 otherwise, with large-size fund as the base
3	medium_fund	Dummy variable =1 if a medium-size fund and 0 otherwise, with large-size fund as the base
4	portfolio_stocks	Percentage of portfolio invested in stocks (1 – 100)
5	portfolio_bonds	Percentage of portfolio invested in bonds (1 – 100)
6	price_earnings	Ratio of price per share to earnings per share
7	price_book	Ratio of price per share to book value per share
8	price_sales	Ratio of price per share to sales per share
9	price_cashflow	Ratio of price per share to cash flow per share
10	label (dependent variable)	1 = winning fund, an ETF whose monthly return is one standard deviation or more above the mean monthly return across all ETFs in the dataset, and 0 = average fund, otherwise

For the first logistic regression, you are asked to use all the independent variables, except for the fund size dummy variables (small_fund and medium_fund). For the second logistic regression, you are asked to use all the independent variables except the fund size continuous variable (net assets).

You use a standard software package (in Python or R) to develop the logistic regression models. Your results are as follows:

Logistic Regression 1

Logistic Regression Results

Model: Logit Pseudo- R^2 : 0.057

Dependent Variable: label

No. Observations: 1,594 Log-Likelihood: -451.66

Df Model: 7 LL-Null: -478.86

Df Residuals: 1,586 LLR P -Value: 1.97E-9

No. Iterations: 10

	Coefficient	Std. Err.	z- Statistic	P -Value > z	[0.025	0.975]
Intercept	-2.0350	0.5221	-3.8979	0.0001	— 3.0583	— 1.0118
net_assets	-0.7667	1.3571	-0.5649	0.5721	— 3.4265	1.8932
portfolio_stocks	-0.0089	0.0051	-1.7550	0.0793	— 0.0188	0.0010
portfolio_bonds	-0.1113	0.0729	-1.5263	0.1269	— 0.2543	0.0316
price_earnings	0.0292	0.0200	1.4647	0.1430	— 0.0099	0.0683
price_book	-0.0390	0.1029	-0.3791	0.7046	— 0.2407	0.1627
price_sales	0.3432	0.0777	4.4160	0.0000	0.1909	0.4956
price_cashflow	-0.0502	0.0363	-1.3805	0.1674	— 0.1214	0.0211

Logistic Regression 2

Logistic Regression Results

Model: Logit Pseudo- R^2 : 0.059

Dependent Variable: label

No. Observations: 1,594 Log-Likelihood: -450.4

Df Model: 8 LL-Null: -478.86

Df Residuals: 1,585 LLR P -Value: 1.87E-09

No. Iterations: 10

	Coefficient	Std. Err.	z-Statistic	P-Value > z 	[0.025	0.975]
Intercept	-1.9589	0.5254	-3.7283	0.0002	-2.9886	0.9291
small_fund	-0.4794	0.3719	-1.2888	0.1975	-1.2083	0.2496
medium_fund	-0.3509	0.2348	-1.4948	0.1350	-0.8111	0.1092
portfolio_stocks	-0.0092	0.0051	-1.8099	0.0703	-0.0191	0.0008
portfolio_bonds	-0.1121	0.0727	-1.5433	0.1228	-0.2546	0.0303
price_earnings	0.0389	0.0211	1.8467	0.0648	0.0024	0.0802
price_book	-0.0803	0.1068	-0.7519	0.4521	-0.2897	0.1291
price_sales	0.3453	0.0796	4.3362	0.0000	0.1892	0.5014
price_cashflow	-0.0510	0.0376	-1.3573	0.1747	-0.1247	0.0227

Identify which one of the following choices is *most likely* to be correct: Logistic regression is the appropriate regression method for your assignment because the:

☐ A. independent variables include dummy variables for small- and medium-size funds.

☒ B. dependent variable is binary rather than continuous.

☐ C. dependent variable is not continuous and the independent variables include dummy variables for small- and medium-size funds.

Feedback

B is correct. Logistic regression is the appropriate regression method because the dependent variable is binary rather than continuous.

6 Multiple Choice 1 / 1 point

Identify which one of the following statements *best describes* the interpretation of an independent variable's slope coefficient in a logistic regression model: The slope coefficient is the change in the:

☐ A. log odds that the event happens per unit change in the independent variable, while all other independent variables increase by one unit.

☐ B. odds that the event happens per unit change in the independent variable, holding all other independent variables constant.

☒ C. log odds that the event happens per unit change in the independent variable, holding all other independent variables constant.

Feedback

C is correct. An independent variable's slope coefficient in a logistic regression model is the change in the log odds that the event happens per unit change in the independent variable, holding all other independent variables constant.

7 Multiple Choice 1 / 1 point

Determine which one of the following statements is *true*. The intercept in these logistic regressions is interpreted as the:

☐ A. probability of the ETF being a winning fund if all independent variables are one.

☒

☐ B. log odds of the ETF being a winning fund if all independent variables are zero.

☐ C. log odds of the ETF being an average fund if all independent variables are zero.

Feedback

B is correct. The intercept in these logistic regressions is interpreted as the log odds of the ETF being a winning fund if all independent variables are zero.

8 Multiple Choice 0 / 1 point

Based on the output in Exhibit 1: Output for Logistic Regression 1, which of the following alternatives is closest to the probability that any ETF will be a winning fund?

Exhibit 1: Output for Logistic Regression 1

Independent Variables	Median	Mean	Standard Deviation
net_assets (\$100B)	0.1687	0.2911	3.5255
portfolio_stocks	99.6400	92.9093	19.6162
portfolio_bonds	0	2.3068	11.9835
price_earnings	14.4750	15.1743	5.4087
price_book	1.7800	2.0711	1.1113
price_sales	1.2600	1.6060	1.1877
price_cashflow	7.0300	7.6489	3.7768

☒

☐ A. 6.75%

Correct Answer: **C. 5.67%**

☐ B. 5.96%

☐ C. 5.67%

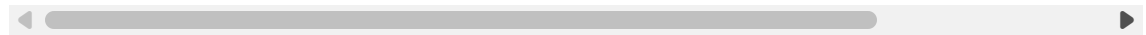
Feedback

C is correct. We calculate the probability that an that an ETF will be a winning fund by using the variable estimates and the average values of the independent variables.

Using the equation for the probability, where we have seven independent variables,

_____.

Using the mean values and coefficient estimates of the independent variables, the probability of the average ETF being a winner is



$P = 5.67\%$.

This implies that for an ETF with the average values of the independent variables, there is a 5.67% probability that it will be a winning ETF.

9 Multiple Choice 0 / 1 point

Based on the output in Exhibit 1: Output for Logistic Regression 1, which of the following alternatives is closest to the change in the probability that an ETF will be a winning fund if its price-to-earnings ratio increases by one unit and all else stays constant?

☒ A.
17.0%

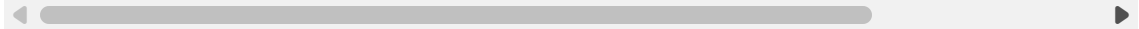
Correct Answer: **B. 0.16%**

☐ B. 0.16%

☐ C. 1.5%

Feedback

B is correct. Keeping all the independent variables' average values fixed and increasing the P/E value by 1, from 15.1743 to 16.1743, yields the new probability.


$$P = 5.82\%.$$

From the previous question, we know that the marginal probability is 5.67%. Therefore, the marginal increase in the probability of an average ETF being a winner went up $5.82\% - 5.67\% = 0.15\%$. A one-unit change in the P/E of an ETF will increase the probability of it becoming a winning fund by 0.15%, and therefore, B is the closest answer.

10 Multiple Choice 1 / 1 point

Assuming that the past regression results seen in Exhibit 1: Output for Logistic Regression 1 accurately describe the future state, which of the following would most reliably lead to a decrease in the probability of success?

- ☐ A. An increase in the price to book ratio.
- ☒ B. A decrease in the price to sales ratio.
- ☐ C. An increase in Net Assets

Feedback

B is correct. The coefficient of price to sales is positive and is statistically different from zero at beyond the 99% level of confidence, so a decrease in this ratio would more likely be associated with a decrease in the probability of success than option A (not distinguishable from zero beyond the 70.5% confidence level) or an increase in Net Assets (which is also not distinguishable from zero beyond a fairly low confidence level of 57.2%).

11 Multiple Choice 1 / 1 point

Identify which one of the following statements about the logistic regression model fit is *most likely* to be correct: Based on the log-likelihood criteria:

- ☒ A. Model 2 has a better fit because it has a higher log-likelihood value.

- ☐ B. Model 2 has a better fit because it has a lower log-likelihood value.
- ☐ C. Model 1 has a better fit because it has a higher log-likelihood value.

Feedback

A is correct. Model 2 has a better fit because it has a higher (less negative) log-likelihood value, -450.40 versus -451.66 , compared to Model 1.

12 Multiple Choice 1 / 1 point

Determine, using only the statistically significant variable estimate(s) in Logistic Regression 2 and the information provided below, which of the following is *closest to* the probability of the Alpha ETF being a winning fund and whether it would be classified as a winning fund.

Alpha ETF variable values:

- `small_fund` = 0.
- `medium_fund` = 0.
- `portfolio_stocks` = 99.3%.
- `portfolio_bonds` = 0.7%.
- `price_earnings` = 25.0.
- `price_book` = 1.1.
- `price_sales` = 4.0.
- `price_cashflow` = 5.7.

Use a significance level of 5% and a probability threshold for being a winner of 65%.

- ☐ A. 27.4%, and the Alpha ETF is not classified as a winning fund
- ☒ B. 36.0%, and the Alpha ETF is not classified as a winning fund

- ☐ C. 82.2%, and the Alpha ETF is classified as a winning fund

Feedback

B is correct. Besides the significant intercept, the only significant (at the 5% level) variable in Logistic Regression 2 is `price_sales`. Using these two factors, the probability of this ETF being a winning fund is calculated to be 35.95%, as follows:

Probability of being a winning fund = 0.3595 =

Because this probability is well below the 65% threshold for being a winner, the Alpha ETF would not be classified as a winning fund.

Vignette

Anna Gabor is an analyst with an investment management firm. She is investigating the impact of leverage on annual ROE in the firm's Eurozone equities fund ("the fund").

She collects data on the 72 stocks in the fund, including company size as an independent variable.

Exhibit 1 shows the results from estimating the model $R_i = b_0 + b_1(\text{Size}_i) + b_2(\text{D/E}_i) + \varepsilon_i$

Where:

- R_i is the annual ROE for company i in the last calendar year in numbers (0.05 is 5%);
- Size_i is the natural log of the market capitalization of company i (millions of Euro) on 31 December;
- D/E_i is the debt/equity ratio of company i on 31 December;
- b_0 is the intercept of the equation;
- b_1, b_2 are the slope coefficients for the independent variables Size_i and D/E_i respectively;
- ε_i is the error term.

Exhibit 1

	Coefficient	Standard Error
Intercept	0.2248	---
Size _i	-0.012	---
D/E _i	-0.0172	0.00986
ANOVA	Degrees of freedom	Sum of squares
Regression	2	0.0960
Residual	69	0.5704

Exhibit 2 shows selected values of Student's t distribution for 69 degrees of freedom.

Exhibit 2
Selected Values of Student's t distribution

Area in Right Tail	t Value (69 degrees of freedom)
0.05	1.6670

0.025

1.9950

Gabor expects to find that both the debt/equity ratio and size are negatively related to ROE. She considers a relationship meaningful when it is statistically significant at the 0.05 level. As part of her analysis Gabor compares the predictions of her model with the actual ROE for two stocks in the portfolio which have the same debt ratio: A and B, which have market capitalizations EUR 100 million and EUR 200 million respectively.

Gabor shares her results with a colleague, Eva Szabo, who warns that the variance of the error terms might not be constant, but positively correlated with D/E and Size. Szabo states that such a flaw would lead to the F- and t-statistics being biased and failing to reject the null hypothesis when it is false.

As an additional goal Gabor investigates the impact of qualitative governance factors. She is considering adding two dummy variables, to the model: D_1 and D_2 . D_1 is 1 for firms with two or more female board members, and otherwise 0. The value of D_2 is 1 if the audit committee of the board includes only independent directors, and otherwise 0. Gabor's hypothesis is that gender diversity is positively related to ROE and that an independent audit committee slows the rate at which increasing leverage affects ROE.

Finally, Gabor wishes to test the hypothesis that whether or not a stock in the fund is listed on a US (as well as European) exchange is determined by the same independent variables as above (including the governance factors). She employs a logistic regression with the dependent variable of A, which takes on a value of 1 for a company listed on a US exchange and otherwise 0. For Tekdat, a company in the fund which does not have gender diversity on its board, the probability, based on the model, of being listed on a US exchange is 0.65. The regression coefficient of D_1 in the logistic model is 0.80 and is statistically significant.

13 Multiple Choice 0 / 1 point

Based on the logistic model and with a diverse board, the probability of Tekdat being listed on a US exchange would be *closest* to:

☐ 0.73.

Incorrect Answer Feedback: Incorrect, because we have not performed the right calculation to reflect the impact of the coefficient of D_1 .

The calculation that leads to this answer does not take into account that the model is based on the natural logarithm of the odds:

P is the probability of being listed on a US exchange. Without a diverse board, for Tekdat, $P = 0.65$

The odds is $\{P/(1-P)\} = 1.8571$

With gender diversity on the board, we calculate effect of an increase in the odds of 0.8:

$1.8571 + 0.8 = 2.6571$

This is incorrect: it is the log odds, not the odds that increases by 0.8

Using these new odds to calculate P_1 i.e. the probability of Tekdat being listed on a US exchange, then

$P_1/(1-P_1) = 2.6571$

And $P_1 = 0.7266$ or approximately 0.73.

☐ 0.81.

Correct Answer Feedback: Correct, because this amount reflects the right interpretation of the coefficient of D_1 in the logistic regression model.

P is the probability of being listed on a US exchange. Without a diverse board, for Tekdat, $P = 0.65$

The log odds = $\ln \{P/(1-P)\} = \ln (1.8571) = 0.6190$

With gender diversity on the board, the log odds increases by 0.8

So is now $0.6190 + 0.8 = 1.4190$

The new odds is the exponent of 1.4190, $\exp (1.4190) = 4.1331$

If P_1 is the probability of Tekdat being listed on a US exchange with this new odds, then

$P_1/(1-P_1) = 4.1331$

And $P_1 = 0.8052$ or approximately 0.81

☒ 0.93.



Incorrect Answer Feedback: Incorrect because this answer is the result of the omission of a step in the calculation, inconsistently applying the method.

P is the probability of being listed on a US exchange. Without a diverse board, for Tekdat, $P = 0.65$

The odds = $P/(1-P) = 1.8571 =$

At this point we omit to take the log odds.

With gender diversity on the board, we increase the odds by 0.8

So is now $1.8571 + 0.8 = 2.6571$

We calculate the new odds as the exponent of 2.6571, $\exp(2.6571) = 14.2555$

If P_1 is the probability of Tekdat being listed on a US exchange with this new odds, then

$P_1/(1-P_1) = 14.2555$

And $P_1 = 0.9344$ or approximately 0.93